

# Public Sentiment and Common Topics on Philippine Agriculture: A Comparative Text Analysis of Reddit Discussions and Online News Articles

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**Abstract**—This study focuses on public sentiments and common topics of agriculture in the Philippines, specially in news platforms and reddit.

**Index Terms**—component, formatting, style, styling, insert

## I. INTRODUCTION

The main intuition in this study is to provide context into what are the common topics and sentiments towards the topic **agriculture** in the Philippines. While it is common knowledge that agriculture is often **unappreciated** and that it is **unsupported**, this study provides a textual analysis on the conversations that occur in online platforms.

The Department of Agriculture (DA) recognizes the fact that the agricultural situation in the Philippines is struggling [1]. These struggles are stem from various challenges that the sector is facing, such as the decreasing number of farmers, harvestable area, and the increasing cost of production [2]. With the continuous rise of population, the demand for food is also increasing. However, the agricultural sector is not able to keep up with this demand due to various challenges such as lacking of government support, climate change, and limited access to new farming technology [2].

### A. Research Questions

- 1) What are the dominant public sentiments toward agriculture in the Philippines across Reddit discussions and news articles?
- 2) What are the most common topics associated with Philippine agriculture in public (Reddit) versus formal (news) discourse?
- 3) How do public sentiments and topics differ between the two sources?
- 4) Is it possible to identify the specific needs or concerns of farmers based on these online discussions?

### B. Scope & Limitations

- 1) Considering the topic utilizes a discussion-based dataset, the study will not take into account the network structure of the data.

- 2) The analysis is limited to text content and does not incorporate multimedia elements such as images or videos, which may also influence discussions or public perception.

## II. METHODOLOGY

TABLE I  
RAW REDDIT DATA SAMPLE

|                       |                                                                             |
|-----------------------|-----------------------------------------------------------------------------|
| <b>post_title</b>     | Sample title about farming                                                  |
| <b>post_url</b>       | https://reddit.com/...                                                      |
| <b>comment</b>        | Agriculture in the Philippines needs reform.                                |
| <b>comment_author</b> | user123                                                                     |
| <b>comment_score</b>  | 12                                                                          |
| <b>post_body</b>      | Full post discussing rice production and its challenges in the Philippines. |
| <b>post_score</b>     | 256                                                                         |
| <b>post_author</b>    | authorA                                                                     |
| <b>created_utc</b>    | 2025-10-23 08:14:00                                                         |

TABLE II  
ONLINE NEWS PLATFORM DATA SAMPLE

|                       |                                                |
|-----------------------|------------------------------------------------|
| <b>title</b>          | Arms to farms: Maguindanao del...              |
| <b>link</b>           | https://www.rappler.com/philippines/...        |
| <b>date_published</b> | 2025-10-24 09:30:00                            |
| <b>author</b>         | Maria Dela Cruz                                |
| <b>text</b>           | COTABATO CITY, Philippines —Maguindanao del... |

### Data Collection & Preprocessing

**Reddit data** is collected from `r/Philippines`, while agriculture-related news articles are taken from **Rappler**. Metadata that are gathered from both sources are summarized in their respective tables ( *See Table I & II* ) The data is cleaned by (i) removing null values, (ii) removing special characters, (iii) normalizing text, and (iv) identifying and removing stopwords.

### Exploratory Data Analysis

At this stage, basic exploratory data analysis (EDA) methods are going to provide us a deeper understanding of what the

data has to offer. (i) Basic Overview and Structure, (ii) Word Frequency Analysis, (iii) Engagement and Metadata Trends, (iv) Common Keywords, (v) Text Comparison (*Reddit vs News, how do they write text?*)

***Natural Language Processing***    ***a. Topic Modelling***  
***b. Sentiment Analysis***

III. RESULTS & DISCUSSION

IV. CONCLUSION

V. POSSIBLE FUTURE WORKS

REFERENCES

REFERENCES

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